Detection of cancer through urine fluorescence

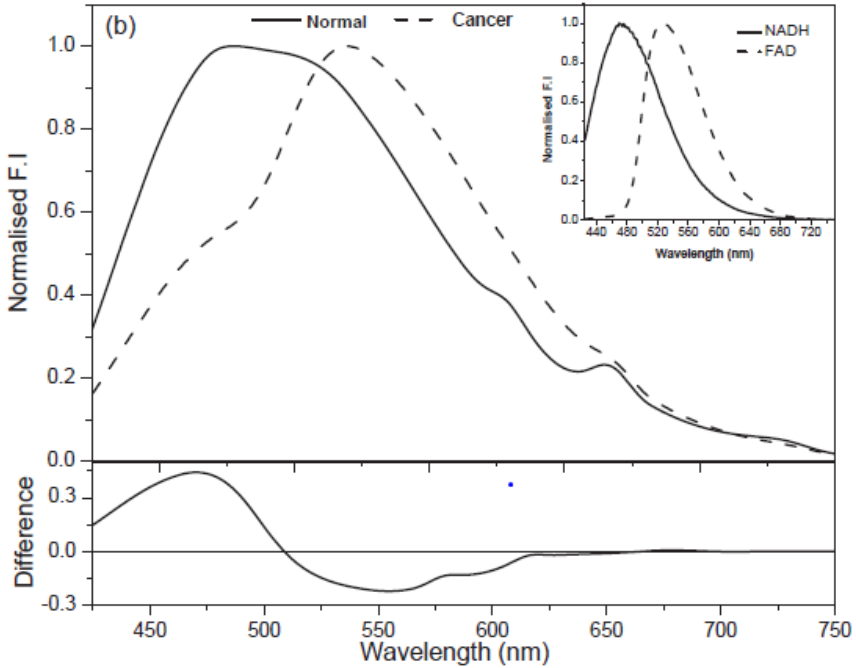
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Problem description

Many metabolites in human urine (e.g. NADH) are native fluorophores

Ratios calculated from fluorescent intensity values of the emission spectra of subjects healthy and with cancer are significantly different.

Fluorescence spectroscopy is a valid diagnostic tool.



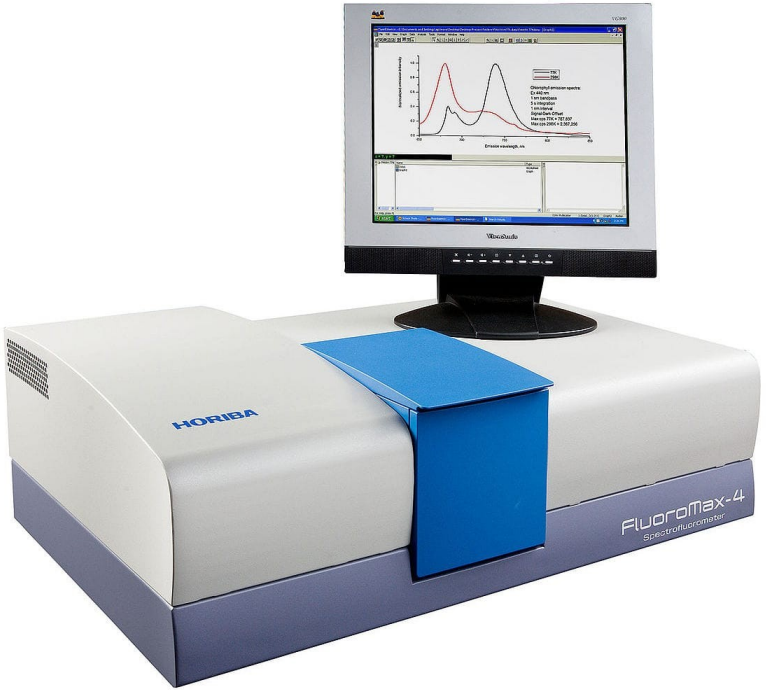
Fluorescence Intensity emission spectra when stimulated at 405nm

Instrumentation project

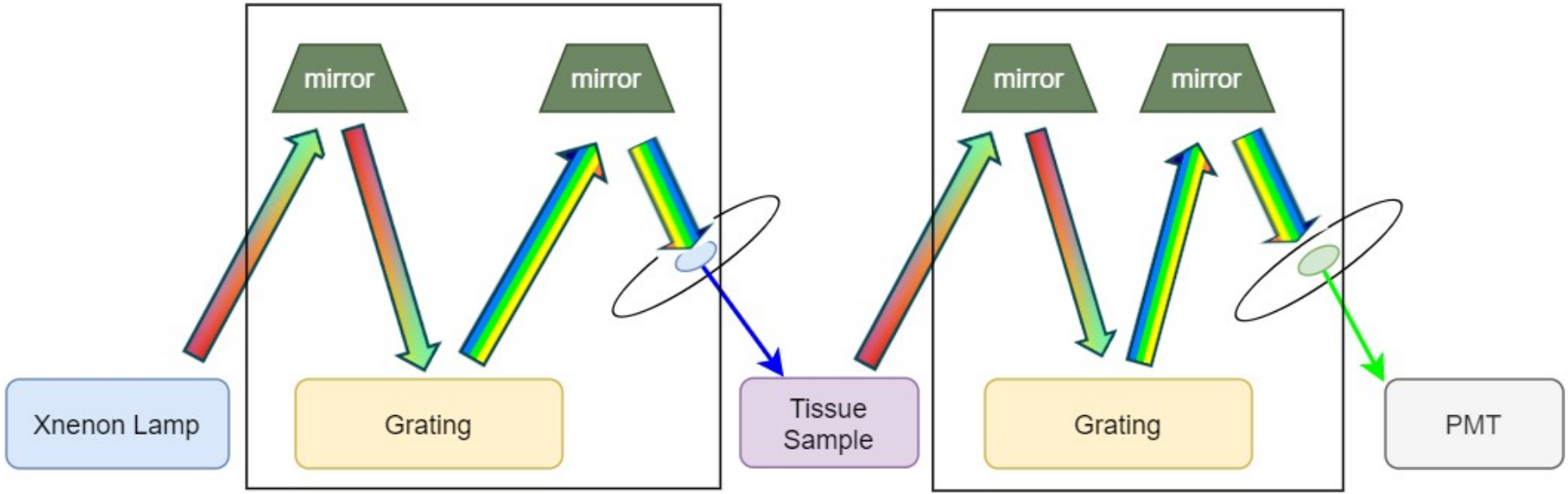
The purpose of this project is to design a spectrofluorometer that detect fluorescence emission from an urine sample.

The instrumentation must enable a choice for the emitted and received wavelength and provide a reliable measure of the emitted intensity.

The sample is excited at 405 nm and the emitted light is measured in the range 460-625 nm.



Block Diagram



Excitation monochromator

Emission monochromator

Component	Model	Price	Important characteristics
ARC LAMP	L2174-02, Hamamatsu SLS205	~500€ 4454,10 €	Xenon Ozone Free 240 - 1200 nm
Excitation Monochromator	GR25-1204	106,23 €	*405nm*
Emission Monochromator	GR25-1205	106,23 €	*470-625nm*
Photomultiplier Tube	PMM02 R928P, Hamamatsu	2424,68 €	

~3140€